

Software Project Interface Prototype

SoundGarden

Version 1

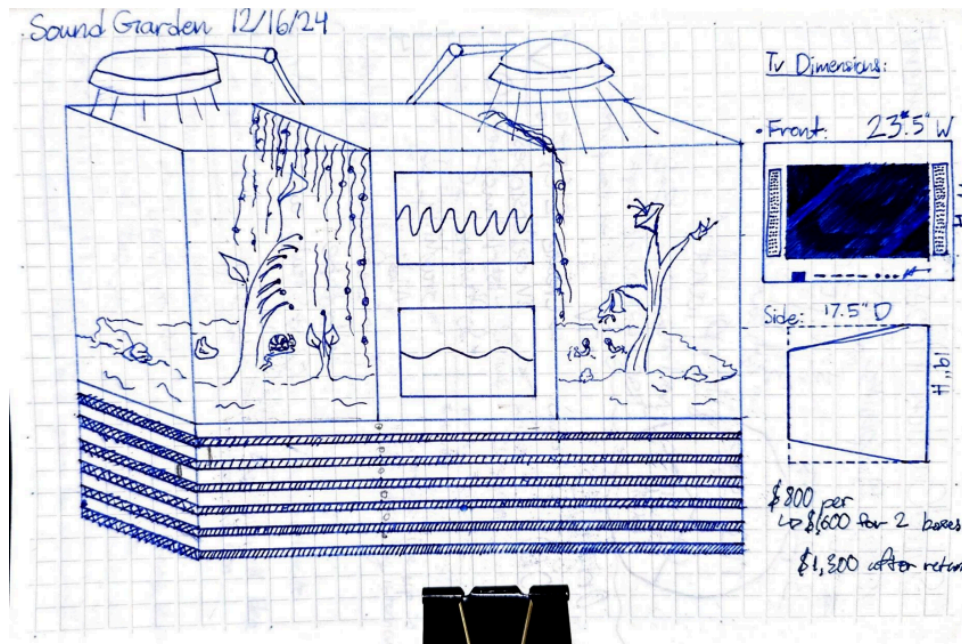


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1 User Interfaces – GUI Versions

Software Application User Interface (UI)

Sound Garden's UI consists of the visualizations that will be displayed on the CRT TVs. These visualizations will use graphs, color, and typography elements to create a display that effectively communicates relevant input/output data to the user (the installation viewer). Sound Garden's UI will not have any buttons or opportunity for user input.

1.1 Oscilloscope Visualizer Interface

1.1.1 Bioelectric Signals Graph (Oscilloscope)

This visualizer will be displayed on the top CRT TV. Using a digital oscilloscope graph, this interface will communicate to viewers the electric activity occurring in the terrarium plants and the direct signals that our program is using as input to create the music. We have a couple options for displaying the planned 8 plant signals. We could alternate between showing the activity from the left then right terrarium sides, as shown by prototype A. This would allow for a less cluttered display, but could lead to confusion if we were to keep playing music from the attenuated terrarium without representing it visually. If we were to only play music from one terrarium at a time, this could dull the musical experience. Alternatively, prototype B displays what it might look like to have all the plant data shown at once in 8 overlapping waves. Prototype A was made with Matplotlib and is an example of a light mode display. Prototype B's graph was made with PyQtGraph and is an example of a dark mode display.

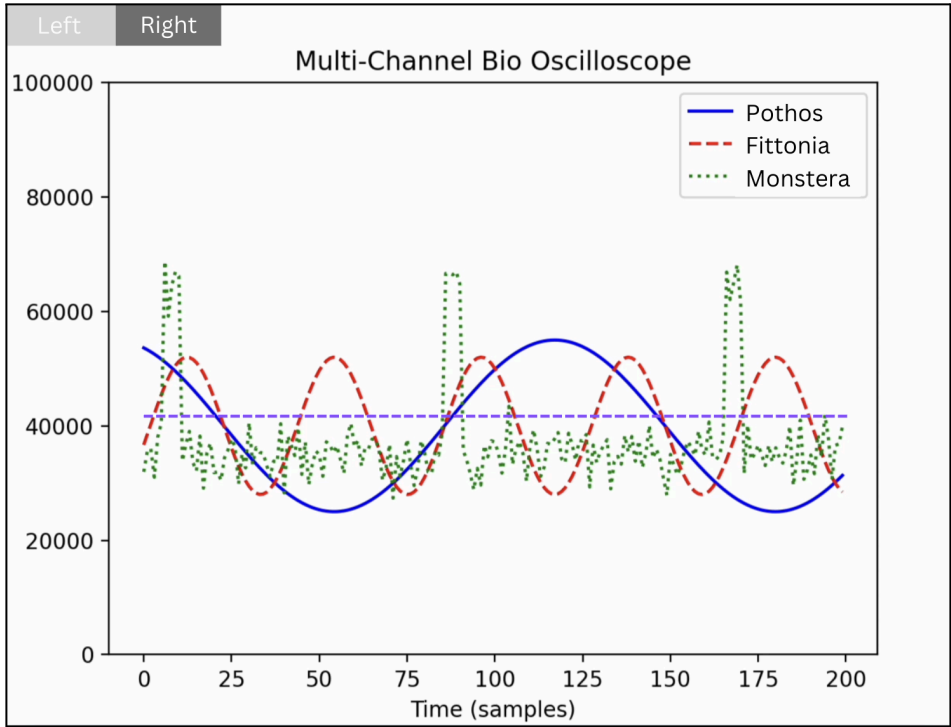
1.1.2 Graph Key

The graph key will allow viewers to understand which wave line corresponds to which specific plant. The waves in the graph will be color coded and composed of different line types as permitted (dashed, dotted, solid, etc.) to create a visually engaging display that is also accessible and colorblind friendly.

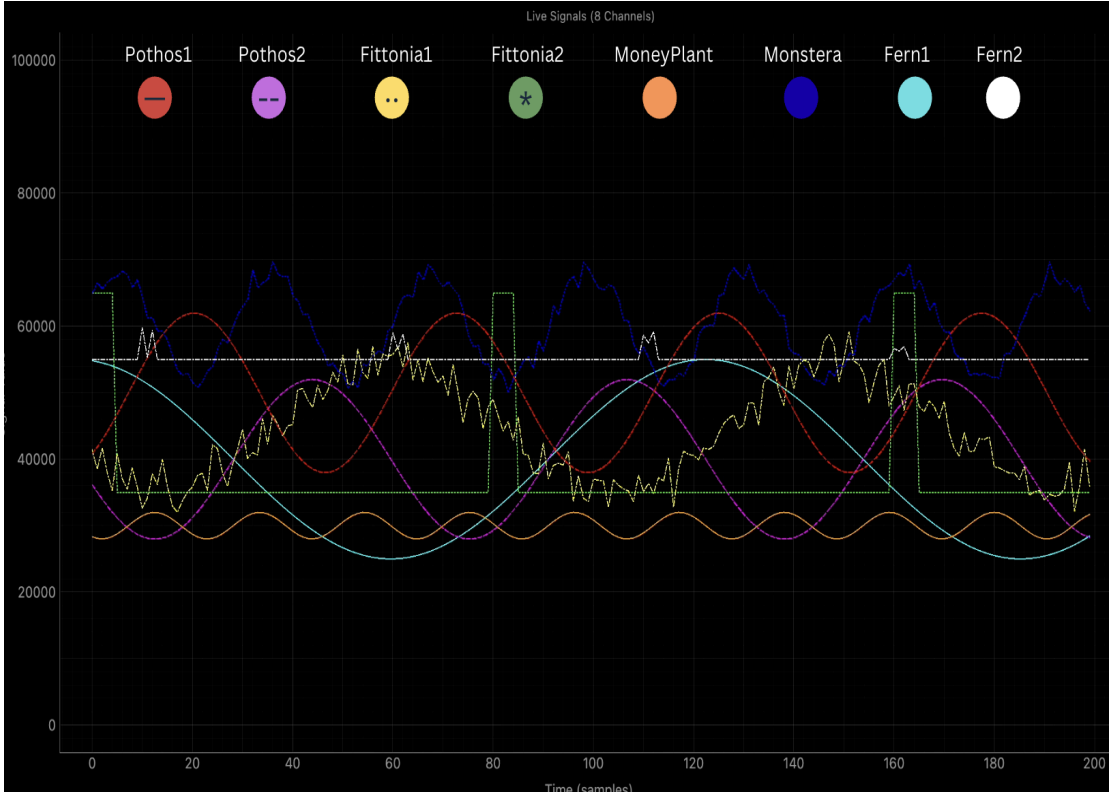
1.1.3 Current Terrarium Label

For the implementation that only shows plants from one terrarium at a time (implementation A), the top left of the display will have a label announcing which side is being focused on for the current time interval. The side that is not currently being displayed will have a more transparent label.

Prototype A:



Prototype B:



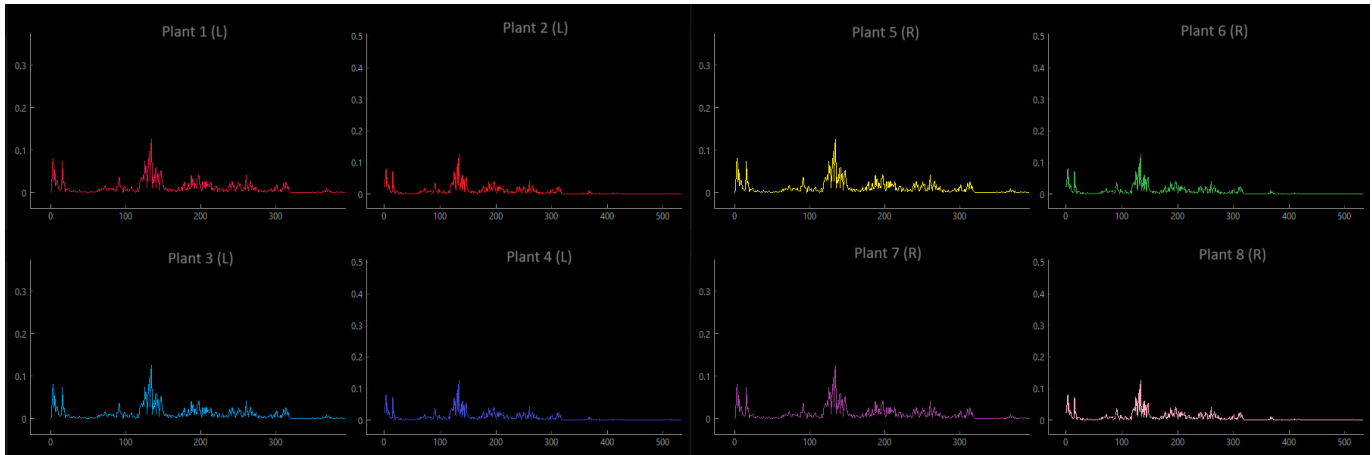
1.2 EQ Graph Visualizer Interface

These visualizations are based on the

<https://github.com/omega0verride/Python-Realtime-Audio-Visualizer?tab=readme-ov-file> output but we don't necessarily have to use that library.

1.2.1 Option A

Option A is we choose to visualize all potential audio outputs at once and we are using 8 plants. Each plant has its own graph, label and color which visualize the music it creates. These colors are not set in stone and the actual colors should ideally be color blind friendly.



1.2.2 Option B

Option B is we overlap two plant signals on top of each other. Each Plant still gets its own color and label but this fits more nicely on the screen. We could also alternatively cycle between which plants are being visualized using this same structure but only showing 4 non-overlapping plants.

